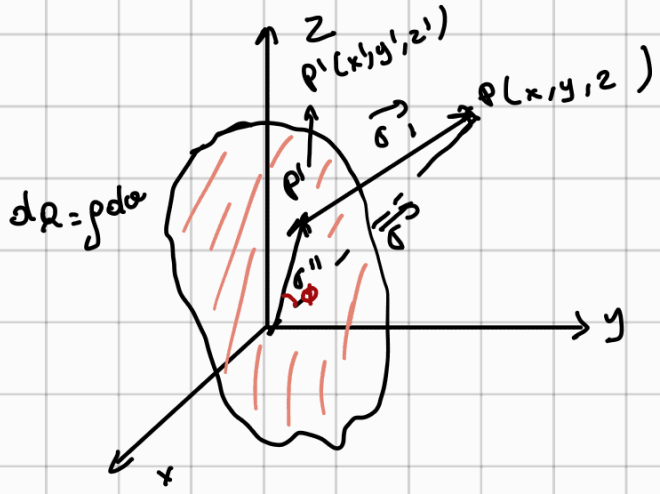


$$r' = \left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{1/2}$$

$$\vec{r}'' = \overrightarrow{OP} = x'\vec{e}_x + y'\vec{e}_y + z'\vec{e}_z$$

$$\vec{r} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z$$

$$\vec{E} = -\text{grad}\, V$$

$$\int_v dv = \frac{1}{4\pi\varepsilon_0}\int_v \frac{dQ}{r'}$$

$$\vec{r}' = \vec{r} - \vec{r}'' = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z - x'\vec{e}_x - y\vec{e}_y - z\vec{e}_z$$

$$V = \frac{1}{4\pi\varepsilon_0}\int_v \frac{\rho}{r'}\,dv$$

$$V = \frac{1}{4\pi\varepsilon_0}\int_{\infty} \frac{\rho}{r'}\,dv \quad \rightarrow \text{coloumb potansiyel}$$

$$V(x,y,z) = \frac{1}{4\pi\varepsilon_0}\int_{x',y',z'=-\infty}^{\infty}\int\int \frac{\rho(x',y',z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\,dx'dy'dz' + V_0$$

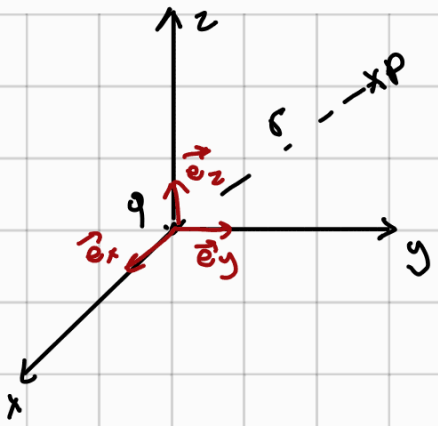
V(x,y,z) = potansiyel fonksiyon

$$\vec{E} = -\text{grad}\, V = -\text{grad}\left\{\frac{1}{4\pi\varepsilon_0}\int_{x',y',z'=-\infty}^{+\infty}\int\int \frac{\rho(x',y',z')}{r'}\,dx'dy'dz' + V_0\right\}$$

$$= -\underset{(x,y,z)}{\text{grad}}\left\{\frac{1}{4\pi\varepsilon_0}\int_{\infty}\int\int \frac{\rho(x',y',z')}{r'}\,dx'dy'dz'\right\} - \text{grad}\,(V_0)$$

$$= \frac{1}{4\pi\varepsilon_0}\int_{x',y',z'=-\infty}^{\infty}\int\int \rho(x',y',z')\,\underbrace{\text{grad}\left(\frac{1}{r'}\right)}_{-\frac{\vec{r}'}{r'^3}}\,dx'dy'dz'$$

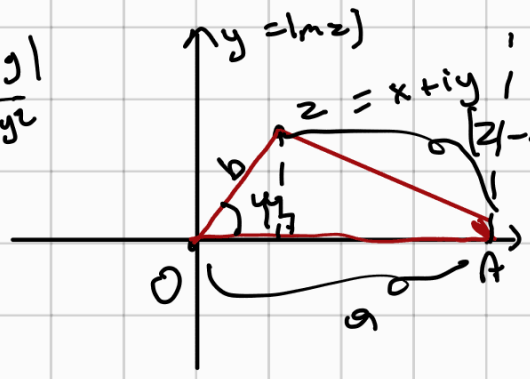
$$\Rightarrow \vec{E}(x,y,z) = \frac{1}{4\pi\varepsilon_0}\int\int\int \frac{\rho(x',y',z')\,\vec{r}'}{r'^3}\,dx'dy'dz'$$



$\vec{e}_x \perp \vec{e}_y \perp \vec{e}_z$

$|\vec{e}_x| = |\vec{e}_y| = |\vec{e}_z| = 1$

$r' = \sqrt{r^2 + r''^2 - 2 \cdot r \cdot r'' \cos\phi}$ cosinus teoremi



$b = |z| = |x + iy| = \sqrt{x^2 + y^2}$

$|z - z_A| = |x + iy - a| = |b\cos\psi + ib\sin\psi - a| = |(b\cos\psi - a) + ib\sin\psi|$

$z_A = a > 0$

$b = |z|$

$\psi = \arctan\left(\frac{y}{x}\right)$ $\arg(z) = \psi$ psi

$\in (-\pi, \pi]$

$z = x + iy = \underbrace{b\cos\psi}_{x} + i\underbrace{b\sin\psi}_{y}$

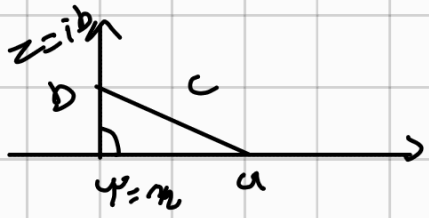
$z = |z|(\cos(\underbrace{\arg z}_{\psi}) + i\sin(\underbrace{\arg z}_{\psi}))$

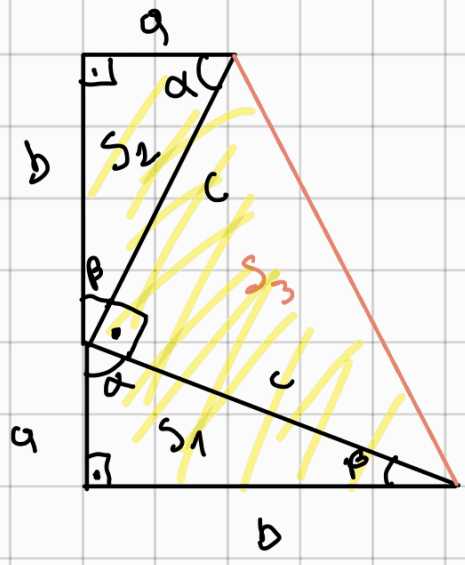
$|z - z_a| = c = \sqrt{(b\cos\psi - a)^2 + (b\sin\psi)^2}$

$c = \sqrt{b^2\cos^2\psi + a^2 - 2ab\cos\psi + b^2\sin^2\psi}$

$c = \sqrt{b^2 + a^2 - 2ab\cos\psi}$

$\psi = \pi/2 \Rightarrow c = \sqrt{b^2 + a^2}$

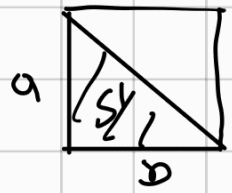




$\alpha + \beta + \gamma = \pi$

$\pi/2$

$\gamma = \pi - \pi/2 = \pi/2$



$S = a \cdot b$

$S' = \frac{S}{2}$

$S_1 = \frac{a \cdot b}{2} = S_2$

$S_3 = \frac{c \cdot c}{2} = \frac{c^2}{2}$

$S_1 + S_2 + S_3 = \frac{ab}{2} + \frac{ab}{2} + \frac{c^2}{2}$

$S_T = ab + \frac{c^2}{2} = \frac{(a+b)(a+b)}{2}$

trapez

$2ab + c^2 = a^2 + b^2 + 2ab$

$c^2 = a^2 + b^2$

OGR

$e^{i\varphi} = \cos\varphi + i\sin\varphi$ ← Eulerov teorem

$e^{-i\varphi} = \cos\varphi - i\sin\varphi$

$e^{i(x+y)} = \cos(x+y) + i\sin(x+y)$

$e^{ix+iy} = e^{ix} \cdot e^{iy} = (\cos x + i\sin x)(\cos y + i\sin y)$

$= \cos(x+y) + i\sin(x+y)$

$(\underbrace{\cos x \cos y - \sin x \sin y}_{\cos(x+y)}) + i(\underbrace{\sin x \cos y + \cos x \sin y}_{\sin(x+y)})$

Sonsuzda küresel cismin noktasal yük gibi davrandığını ispatlayalım

$$r' = \sqrt{r^2 + r''^2 - 2r\,r''\cos\psi}$$

$$\frac{1}{4\pi\varepsilon_0}\iiint_{\upsilon}\frac{\rho(x',y',z')}{(r^2+r''^2-2rr''\cos\psi)^{1/2}}\,d\upsilon$$

$$= \frac{1}{4\pi\varepsilon_0}\int_{x'}\int_{y'}\int_{z'(\upsilon)}\frac{\rho(x',y',z')}{r\left[1-\frac{2}{r}r''\cos\psi+\frac{r''^2}{r^2}\right]^{1/2}}\,dx'dy'dz'$$

$$= \frac{1}{4\pi\varepsilon_0 r}\int \rho(x',y',z')\left[1-\underbrace{\frac{2}{r}r''\cos\psi+\left(\frac{r''}{r}\right)^2}_{<<1}\right]^{-1/2}d\upsilon$$

$r >> r''$ $\quad \frac{r''}{r} << 1 \quad \left(\frac{r''}{r}\right)^2 <<< 1$

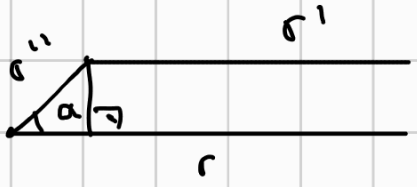
$$(1-x)^{-1/2} \approx 1+\frac{1}{2}x-\frac{1}{8}?x^2-\cdots \qquad |x|<1$$

$$= \frac{1}{4\pi\varepsilon_0 r}\int \rho(x',y',z')\left[\underbrace{1+\frac{r''}{r}\cos\psi-\frac{r''^2}{2r^2}}_{1}\right]$$

$$\approx \frac{1}{4\pi\varepsilon_0}\underbrace{\int_{\upsilon}\rho(x',y',z')\,d\upsilon}_{Q \text{ dersek}} = \frac{Q}{4\pi\varepsilon_0 r}$$

$$\frac{1}{r'} = \frac{1}{r\sqrt{1+\underbrace{\frac{-2r''\cos\psi+\left(\frac{r''}{r}\right)^2}{r}}_{0}}} = \frac{1}{r\cdot\sqrt{1}} \simeq \frac{1}{r}$$

$r' \to \infty$



$$r = r' + r'' \cos\psi$$

$$r' \simeq r - r'' \cos\psi \simeq r\left[1 - \frac{r''}{r}\cos\psi\right]$$

$$1/r' \simeq \frac{1}{r}\left[1 - \frac{r''}{r}\cos\psi\right]^{-1}$$

$$\approx \frac{1}{r}\left(1 + \frac{r''}{r}\cos\psi\right) = 1/r$$

$$e^{ikr} \approx e^{ikr'} = e^{ik(r - r''\cos\psi)}$$

$$= e^{ikr} \cdot e^{-ikr''\cos\psi}$$

$f(x,y,z) = sbt \Rightarrow$ yüzey ailesi verir

$r = \sqrt{x^2+y^2+z^2} = sabit$

Ozaman $V(x,y,z) = sbt \Rightarrow$ Eş potansiyel yüzeyleri

$\vec{E} = -\text{grad}\, V$

$f(x,y,z)$ $z = z(s)$

$x = x(s)$ $y = y(s)$

eğrisi

$$f(C) = f(x(s), y(s), z(s))$$

$$\Rightarrow \frac{\partial f}{\partial C} \hat{=} \frac{\partial f}{\partial s}$$

$$= \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial s}$$

$f(x,y,z)$ fonksiyonun parametrik denklemleri $x = x(s)$, $y = y(s)$ ve $z = z(s)$ olan bir C eğrisi boyunca türevi

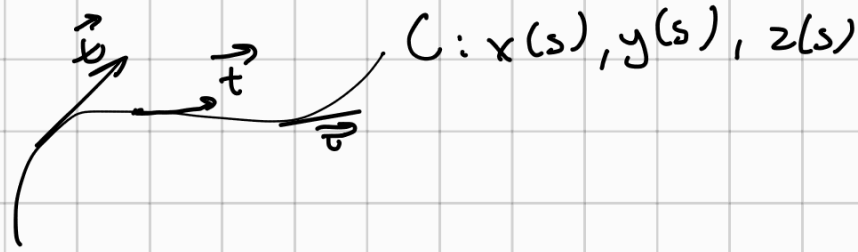
$$\frac{\partial f}{\partial C} = \underbrace{\left(\frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z\right)}_{\text{grad } f = \vec{\nabla} f} \underbrace{\left(\frac{\partial x}{\partial s}\vec{e}_x + \frac{\partial y}{\partial s}\vec{e}_y + \frac{\partial z}{\partial s}\vec{e}_z\right)}_{\vec{t}}$$

$\vec{t}$ → teğet denklem

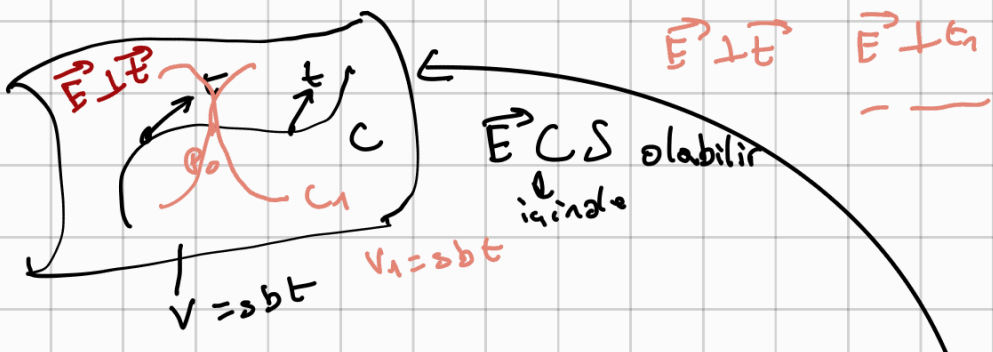
$$\frac{\partial f}{\partial C} = \text{grad } f \cdot \vec{t}$$

$$\frac{\partial V}{\partial C} = \text{grad } V \cdot \vec{t} = -\vec{E} \cdot \vec{t}$$

$$\vec{E} \cdot \vec{t} = -\frac{\partial V}{\partial C}$$



Şayet V = sabit ise $\Rightarrow -\frac{\partial V}{\partial C} = 0$ olur



$\vec{E} \perp \vec{t}$ $\vec{E} \perp t_1$ $\vec{E} \perp t_2$ — — t_n

$$V(C) = sbt$$

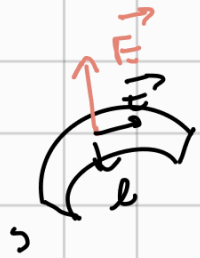
$$\frac{\partial V}{\partial C} = \frac{\partial sbt}{\partial C} = 0$$

$\Rightarrow$ Eş potansiyel yüzey üzerinde

$$\vec{E} \cdot \vec{t} = 0 \qquad \vec{E} \perp \vec{t}$$

$\Rightarrow$ Eş pot yüzeyi $\Rightarrow \frac{\partial V}{\partial C_1} = \vec{E} \cdot \vec{t}_1$

$\Rightarrow \vec{E} \perp B$ (sürtedik)



$\vec{E} \parallel \vec{t}_l$ l'alan çizgisidir

$\vec{t}_l = \frac{dx}{ds}\vec{e}_x + \frac{dy}{ds}\vec{e}_y + \frac{dz}{ds}\vec{e}_z$

$$\frac{E_x}{\frac{dx}{ds}} = \frac{E_y}{\frac{dy}{ds}} = \frac{E_z}{\frac{dz}{ds}}$$

$\vec{E} = E_x \vec{e}_x + E_y \vec{e}_y + E_z \vec{e}_z$

$\vec{t}_1 = \frac{dx}{ds}\vec{e}_x + \frac{dy}{ds}\vec{e}_y + \frac{dz}{ds}\vec{e}_z$

$\vec{E} \parallel \vec{t}_1 \Rightarrow \frac{E_x}{\frac{dx}{ds}} = \frac{E_y}{\frac{dy}{ds}} = \frac{E_z}{\frac{dz}{dz}}$

Alan çizgileri

$$\frac{E_x}{dx} = \frac{E_y}{dy} = \frac{E_z}{dz} = \text{sabit}$$

$\vec{a} \parallel \vec{b}$

$\vec{b} = k\vec{a}$

$k = \frac{b_x}{a_x} = \frac{b_y}{a_y} = \frac{b_z}{a_z} = \text{sbt}$